

Project Brief: QuickBite Sales & Menu Performance Analytics on Amazon SageMaker Unified Studio

Prepared for: Workshop build team

Derived from: QuickBite Glue-native project brief (fast-food franchising client demo)

Purpose: Give developers a single, actionable reference for what gets built, in what order, by whom, and how it's verified.

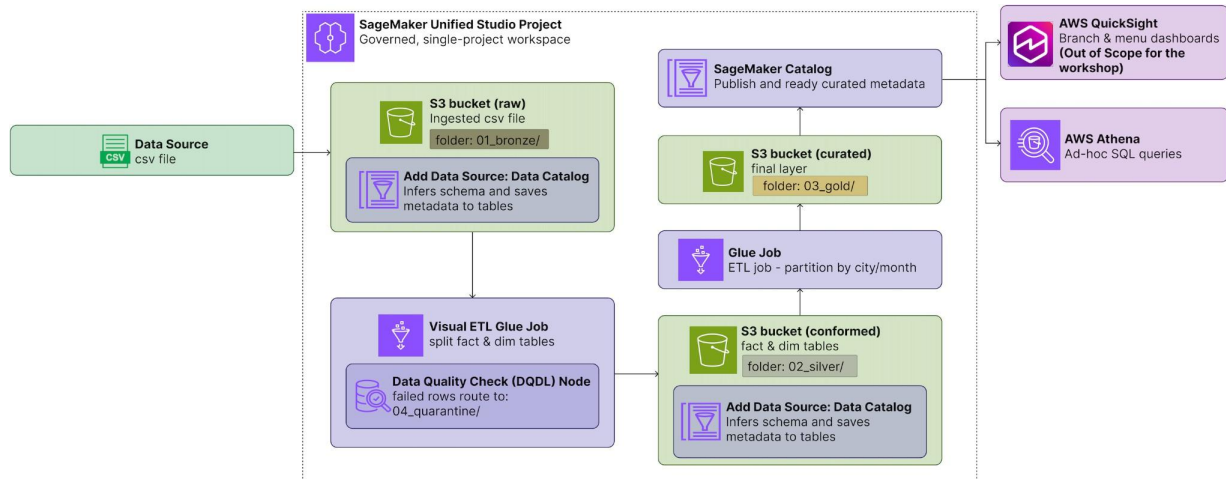
1. Scope note: prototype vs. production

This brief targets a working prototype, not a production-hardened deployment. The following are deliberately deferred, not overlooked, so this reads as a conscious tradeoff rather than a corner quietly cut.

Deferred item	What we do instead for the prototype	Why it's safe to defer here
Fine-grained vs. compatibility permission-mode testing	Use compatibility mode only, for everything	Fine-grained (row/column-level Lake Formation) access matters once multiple teams with different data sensitivity share the project, not yet the case for a single-team prototype
Full least-privilege IAM role separation (Manage access vs. Query execution roles built and scoped independently)	Accept the default roles that quick setup creates for you	Quick setup's defaults are broader than a hand-scoped policy, but this is an internal prototype, not customer-facing or handling regulated data
Per-project cost allocation tagging	Skip; monitor spend informally via the Billing console	Cost isolation matters once multiple teams/projects share an account; not a blocker for a single short-lived prototype
Manual VPC/domain setup	Use quick setup, which auto-provisions the VPC, default roles, and default encryption in one step	Quick setup's auto-created VPC template is explicitly not recommended for production, but it's the fastest path to a working environment for a prototype

Before this goes anywhere near the client or production data, revisit all four rows above.

2. High-Level Architecture



The raw csv file lands in a SageMaker Unified Studio project workspace. Ingested in an S3 bronze bucket, adding a data source to infer the schema and save it in the data catalog as tables. A Visual ETL job processed to conduct data quality checks and split data into dimension and fact tables in parquet format, saved in an S3 conformed bucket. Another Glue Script job used for partitioning the data by city/month and saving the exported Parquet data to S3 curated bucket. Created SageMaker Catalog for publish ready curated metadata that will be used to query data in AWS Athena and QuickSight

Resources Used in SageMaker Unified Studio Workspace

Resource Name	Resource
01_bronze/	S3 bucket prefix (raw)
qbs_smus_bronze	Data catalog - database
sales_raw	Data catalog - table
qbs-bronze-to-silver	Visual ETL job
dqdl-ruleset	Evaluate data quality node
02_silver/	S3 bucket (conformed)
qbs_smus_silver	Data catalog - database
dim_branch	Data catalog - table
dim_product	Data catalog - table
fact_sales	Data catalog - table
qbs-silver-to-gold	Data processing job script

03_gold/	S3 bucket prefix (curated)
qbs_smus_gold	Data catalog - database
branch_performance	Data catalog - table
channel_growth_correlation	Data catalog - table
channel_mix	Data catalog - table
menu_quadrant	Data catalog - table
menu_rank_volatility	Data catalog - table
04_quarantine	S3 bucket prefix (quarantined data)
qbs_smus_quarantine	Data catalog - database
quarantined_data	Data catalog - table

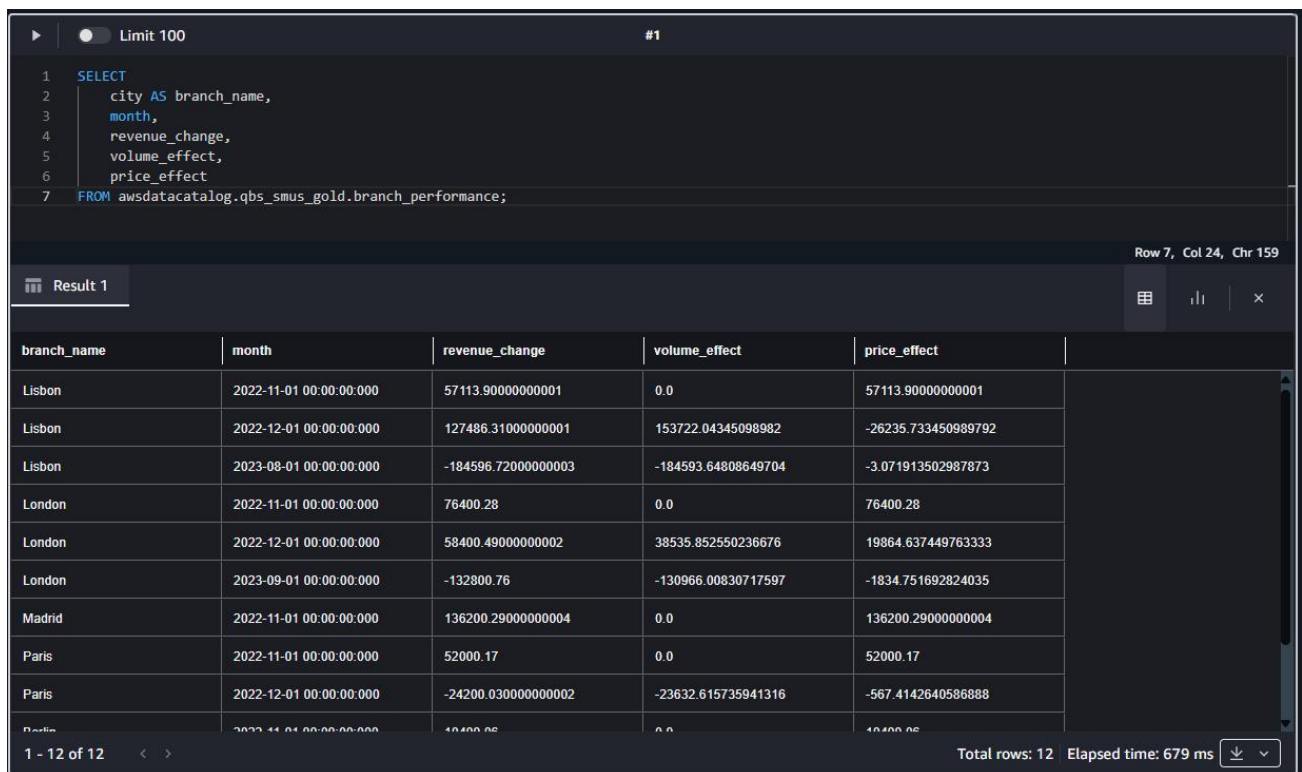
3. Business Question Formulas

- **Business Question 1:** Branch performance: is it price or volume?

```
monthly_revenue = SUM(total_amount) GROUP BY branch_id, month
monthly_qty     = SUM(quantity)   GROUP BY branch_id, month
avg_price       = monthly_revenue / monthly_qty

revenue_change = monthly_revenue - prior_month_revenue    (per branch)
volume_effect  = qty_change * prior_month_avg_price
price_effect   = price_change * current_month_qty
-- volume_effect + price_effect ~= revenue_change (small interaction term ignored)
```

This answers "is it price or volume," not just "revenue grew X%."



Result 1

branch_name	month	revenue_change	volume_effect	price_effect
Lisbon	2022-11-01 00:00:00.000	57113.900000000001	0.0	57113.900000000001
Lisbon	2022-12-01 00:00:00.000	127486.310000000001	153722.04345098982	-26235.733450989792
Lisbon	2023-08-01 00:00:00.000	-184596.720000000003	-184593.64808649704	-3.071913502987873
London	2022-11-01 00:00:00.000	76400.28	0.0	76400.28
London	2022-12-01 00:00:00.000	58400.490000000002	38535.852550236676	19864.637449763333
London	2023-09-01 00:00:00.000	-132800.76	-130966.00830717597	-1834.751692824035
Madrid	2022-11-01 00:00:00.000	136200.290000000004	0.0	136200.290000000004
Paris	2022-11-01 00:00:00.000	52000.17	0.0	52000.17
Paris	2022-12-01 00:00:00.000	-24200.030000000002	-23632.615735941316	-567.4142640586888
Paris	2023-01-01 00:00:00.000	40400.05	0.0	40400.05

1 - 12 of 12 Total rows: 12 Elapsed time: 679 ms

- **Business Question 2:** Menu engineering: which products win where, and does it vary by branch? (Amazon Q-assisted)

```
revenue_tier = NTILE(2) of SUM(total_amount) PARTITION BY branch_id, product_id
volume_tier  = NTILE(2) of SUM(quantity)     PARTITION BY branch_id, product_id
quadrant     = core (high rev, high vol) | premium/niche (high rev, low vol)
              | value driver (low rev, high vol) | underperformer (low rev, low vol)

revenue_rank_per_branch = RANK() of product BY SUM(total_amount) PARTITION BY branch_id
rank_volatility          = STDDEV(revenue_rank_per_branch) GROUP BY product_id
-- high rank_volatility = "star in one city, underperformer in another"
```

Menu Quadrant

▶ Limit 100 #1

```
1 SELECT
2   city AS branch_name,
3   product_id,
4   revenue,
5   volume,
6   revenue_tier,
7   volume_tier,
8   quadrant
9 FROM awsdatacatalog.qbs_smus_gold.menu_quadrant;
```

Row 9, Col 25, Chr 178

Result 1

Row 9, Col 25, Chr 178

branch_name	product_id	revenue	volume	revenue_tier	volume_tier	quadrant
Lisbon	41a209c8e250094e010886fc4...	15000.0	3006.0000000000001	1	1	underperformer
Lisbon	4ca8d115d3f170926ed54fe0d...	36643.4200000000006	3297.5500000000001	1	1	underperformer
Lisbon	7990f92cc9e01087bfa36a0fb...	108799.839999999998	8375.6600000000002	2	1	premiumniche
Lisbon	db720b3728e0536963d88414...	47674.339999999998	9684.7900000000003	2	2	core
Lisbon	e0e8b1de3583536cb5f8e70d...	33599.999999999999	11389.8400000000002	1	2	value driver
London	41a209c8e250094e010886fc4...	14000.0	2805.6000000000001	1	1	underperformer
London	4ca8d115d3f170926ed54fe0d...	32000.7999999999992	3216.1600000000008	1	1	underperformer
London	7990f92cc9e01087bfa36a0fb...	101800.29	7836.8200000000001	2	1	premiumniche
London	e0e8b1de3583536cb5f8e70d...	28800.059999999999	9762.7400000000002	1	2	value driver

1 - 25 of 25

Total rows: 25 Elapsed time: 856 ms

Rank Volatility

▶ Limit 100 #1

```
1 SELECT
2   product_name,
3   rank_volatility
4 FROM awsdatacatalog.qbs_smus_gold.menu_rank_volatility;
```

Row 4, Col 24, Chr 104

Result 1

Row 4, Col 24, Chr 104

product_name	rank_volatility
Burgers	0.0
Fries	0.8944271909999159
Sides & Other	0.0
Chicken Sandwiches	0.4472135954999579
Beverages	0.8944271909999159

1 - 5 of 5

Total rows: 5 Elapsed time: 732 ms

No cost/margin data exists in this dataset, this is a revenue/volume proxy for menu engineering, not true margin-based analysis; say that plainly if asked.

Attendees describe the rank-volatility check in natural language to Amazon Q inside the Visual ETL authoring panel (e.g., "find products whose revenue rank varies the most across branches"); Q drafts the window-function logic, and the attendee checks it against the quadrant/volatility definitions above before trusting the output.

- **Business Question 3:** Channel behavior: does channel mix track with branch performance?

```
channel_revenue_share = SUM(total_amount) PARTITION BY branch_id, purchase_type  
                        / SUM(total_amount) PARTITION BY branch_id
```

```
channel_growth_correlation = CORR(pct_online_revenue, revenue_growth_rate)  
-- one row per branch, revenue_growth_rate from BQ1
```

With only 5 branches this correlation is directional at best, not statistically meaningful, say that rather than overselling a 5-point correlation. It illustrates the method to apply once the client has 50+ branches.

Channel Mix

The screenshot shows a SQL query editor with a query to retrieve channel mix data. The query is as follows:

```
1 SELECT  
2   city AS branch_name,  
3   purchase_type,  
4   orders,  
5   revenue,  
6   pct_of_branch_revenue  
7 FROM awsdatacatalog.qbs_smus_gold.channel_mix;
```

The results are displayed in a table with the following columns: branch_name, purchase_type, orders, revenue, and pct_of_branch_revenue. The table contains 14 rows of data.

branch_name	purchase_type	orders	revenue	pct_of_branch_revenue
Lisbon	Drive-thru	8	20200.1	0.08356900780083866
Lisbon	In-store	35	128913.70999999999	0.5333236388248105
Lisbon	Online	33	92603.79	0.3831073533743509
London	Online	65	184600.95999999996	0.8658538564489313
London	Drive-thru	11	28600.1	0.13414614355106866
Madrid	In-store	25	75400.18000000001	0.5535977933674003
Madrid	Online	1	2199.99	0.01615260877932051
Madrid	Drive-thru	20	58600.119999999995	0.4302495978532791
Paris	In-store	6	16200.05	0.20300735673833847
Paris	Drive-thru	43	25800.43	0.4482313148706030

Channel Growth Correlation

The screenshot shows a SQL query editor with a query to retrieve channel growth correlation. The query is as follows:

```
1 SELECT  
2   channel_growth_correlation  
3 FROM awsdatacatalog.qbs_smus_gold.channel_growth_correlation;
```

The results are displayed in a table with the following columns: channel_growth_correlation. The table contains 1 row of data.

channel_growth_correlation
-0.7456564615568904

4. Infrastructure Build Tasks

Reduced scope per Section 1: quick setup replaces manual VPC/domain creation, default roles replace hand-scoped least-privilege roles, cost tagging is skipped, and fine-grained permission mode is not configured or tested for this prototype.

#	Task	Prerequisite	Stakeholder decision needed?	Status
I1	Decide domain type (IAM-based vs. IAM Identity Center-based)	None	Use SSO log in	completed
I2	Create the SageMaker Unified Studio domain using quick setup, which auto-provisions the VPC, default IAM roles, and default encryption in one step	Task I1	Yes — confirm domain name and whether Bedrock model access should be granted for the Amazon Q demo	completed
I3	Create S3 bucket (`smus/01_bronze`, `smus/02_silver`, `smus/03_gold` & `smus/04_quarantine`) with public access blocked	Task I2	Yes — confirm naming convention Optional: retention/lifecycle policy for the rejected-records bucket	Completed
I4	Configure IAM Identity Center integration: permission set (QuickBite-Workshop-Attendee), directory group (quickbite-etl-team)	Task I1 (only if IDC-based domain chosen)	Yes — confirm which directory group(s) attendees belong to, and whether QuickSight will share the same IDC instance	Completed
I5	Create the workshop project; assign instructor(s) as Project Owner, attendees as Project Contributor	Task I2	No — mechanical once domain exists	completed

I6	Verify the default Glue ETL compute connection exists and use it as-is (compatibility mode only; fine-grained mode deferred per Section 1)	Task I5	No	completed
I7	Create table `AwsDataCatalog.qbs_smus_bronze` in unified studio data catalogs, pointing to the raw S3 prefix.	Task I5	No	completed
I8	Confirm QuickSight blueprint prerequisites: QuickSight account in the same AWS account and same IAM Identity Center instance as the domain	Task I4	Yes — confirm QuickSight licensing (Author/Admin vs. Reader) for attendees	completed

5. Data Engineering Build Tasks

#	Task	Prerequisite	Assigned Dev	Status
D1	Upload Sales-Data-Analysis.csv to the unified studio's project S3 bucket (`bronze/sales_raw`)	Task I3, I7	DE team	Done ▾
D2	Build a Visual ETL source node reading raw data directly from Amazon S3. Service: Visual ETL, backed by AWS Glue 5.1 serverless Spark.	Task D1	DE team	Done ▾
D3	Investigate the fractional Quantity pattern (values like 573.87) before building any logic that depends on it, profile via a notebook or the Visual ETL data preview, and document whether it's a legitimate weighted figure or a data-quality artifact Note:confirm how fractional Quantity should be treated once the pattern is understood	Task D2	DE team	Done ▾

D4	Build the star-schema split: derive dim_product, dim_branch, and fact_sales (with total_amount computed as unit_price x quantity). Service: Visual ETL Derived column and split/mapping transform nodes on AWS Glue Spark	Task D3	DE team	Done ▾
D5	Add an Evaluate Data Quality transform node with the DQDL ruleset: Order ID not null, Price > 0, Quantity > 0, City in known branch list. Service: Visual ETL's Evaluate Data Quality node (same DQDL syntax as a native Glue job)	Task D4	DE team	Done ▾
D6	Implement the negative-test handling from Section 7 directly in the flow: de-duplicate duplicate Order IDs, unrecognized `Products` and unknown `City` values sent to quarantine prefix for review and route unparseable dates to an errors/ prefix instead of failing silently. Service: Visual ETL filter/branch transform nodes, with a second output branch to the quarantine bucket	Task D4 (Optional)	DE team	Done ▾
D7	Build the Business Question 1 flow (branch performance: price vs. volume) using the formula in Section 3	Tasks D4-D6	Jem	Done ▾
D8	Build the Business Question 2 flow (menu engineering quadrant + rank volatility) using Amazon Q: describe the rank-volatility check in natural language inside the Visual ETL authoring panel and let Q draft the window-function logic Reference: https://dzd-beaiy3asjwltcx.sagemaker.	Task I2 decision on Bedrock access; Tasks D4-D6 Note: done in notebook with assistance with Q	Jem	Done ▾

	ap-southeast-1.on.aws/projects/bx6wr6wcd35wh/notebooks/home			
D9	Review the Amazon Q-generated flow against the Section 3 quadrant/volatility formulas; adjust manually if it doesn't match	Task D8		To do ▾
D10	Build the Business Question 3 flow (channel behavior and its correlation with branch performance) using the formula in Section 3	Tasks D4-D6		To do ▾
D11	Write the fact_sales table and both dimension tables directly to the curated S3 bucket in Parquet format using AWS Glue 5.1.	Tasks D7, D9, D10	DE team	Done ▾
D12	Register fact_sales, dim_product, and dim_branch to SageMaker Catalog so they are discoverable domain-wide.	Task D11	DE team	Done ▾
D13	Enrich the published assets with column-level business metadata and glossary terms (e.g., define "core product," "value driver," and note the fractional-Quantity caveat directly on the asset)	Task D12		To do ▾
D14	Notebook capability demo: launch a JupyterLab notebook from the project's Build menu, attach it to project compute, and run a small EDA against fact_sales (see Section 6)	Task D11	Jem	Done ▾
D15	Build the orchestration workflow (crawl, transform, data-quality check, publish) using the visual workflow builder	Task D11 (Optional)	DE team	Done ▾
D16	Schedule the workflow via EventBridge Scheduler, directly from the authoring UI	Task D15 (Optional)		Not Implem... ▾

	Note: confirm schedule cadence appropriate for a workshop vs. production cadence			
D17	Launch QuickSight from the project and build a dashboard (branch performance, channel mix by branch)	Tasks D12, I8 (Optional)		Not Implem... ▾
D18	Publish the dashboard to SageMaker Catalog and share it with the workshop cohort group	Task D17 (Optional)		Not Implem... ▾

6. Notebook Capability Demo (Task D14 detail)

A short, deliberate detour into code-first work, meant to show attendees that Visual ETL isn't the only way to touch the data, and it doubles as the fastest way to eyeball the fractional-Quantity pattern from Task D3.

Setup:

- From the project's Build menu, choose JupyterLab.
- Create a new notebook and attach it to a compute: either a Python 3 kernel for lightweight pandas/awswrangler work, or a Glue PySpark kernel (project's Glue interactive session) if the curated tables are large enough to want Spark.
- Confirm the notebook is running inside the project context, so it automatically inherits the project's IAM permissions (no separate credential setup needed).

Access the data:

```
# Python 3 kernel, query via Athena using awswrangler
import awswrangler as wr

df = wr.athena.read_sql_query(
    sql="SELECT * FROM curated_db.fact_sales LIMIT 5000",
    database="curated_db"
)
```

Small EDA: (enough to demonstrate the capability, not a full analysis)

```
# Null counts per column
df.isna().sum()

# Fractional-Quantity check from Task D3
df["quantity"].apply(lambda x: x % 1 != 0).value_counts()
df["quantity"].describe()
```

```
# Top 5 products by revenue
df.groupby("product_id")["total_amount"].sum().sort_values(ascending=False).head(5)
```

The point of this task is narrow on purpose: prove that the same governed data reachable from Visual ETL is one click away in a notebook, with no separate connection string or credential to manage.

7. Testing Plan Checklist

Reduced scope per Section 1: no fine-grained/compatibility permission-mode comparison test, no IAM least-privilege separation test, and no cost-tagging test for this prototype.

- ☐ VPC/network: Glue ETL job reads S3 raw-zone data without requiring a NAT gateway (quick setup's auto-created VPC endpoint routing confirmed)
- ☐ Encryption at rest: default encryption on the S3 buckets is active (default AWS-owned key from quick setup is acceptable for this prototype)
- ☐ Domain/project isolation: an attendee assigned only to Project A cannot view Project B's assets
- ☐ Negative test — negative or zero quantity: a seeded row with Quantity = -5 is rejected by the DQDL rule
- ☐ Negative test — missing price: a seeded row with null Price is quarantined, not treated as 0
- ☐ Negative test — fractional quantity: a value like 573.87 is flagged for review, not silently folded into headline totals until the pattern is understood
- ☐ Negative test — duplicate Order ID: a seeded duplicate is deduped, not double-counted
- ☐ Negative test — unrecognized product: a seeded Product value outside the 5 known categories causes dim_product to auto-extend without the flow crashing
- ☐ Negative test — unknown branch/city: a seeded City not among the 5 known branches is flagged for review, not silently merged in
- ☐ Negative test — date format inconsistency: a row with a different date format fails gracefully to an errors/ prefix, not a silent bad date
- ☐ Business Question 1 accuracy: Visual ETL aggregate matches an independent Athena SQL query on raw data, within rounding tolerance
- ☐ Business Question 2 (Amazon Q) accuracy: Q-generated flow output matches a manually authored equivalent query, or discrepancy is documented and explained
- ☐ Catalog publish: tables published in Task D12 are discoverable and searchable from a different project, and the subscription request/approval flow works end to end
- ☐ Notebook access: a notebook attached to the project's compute can query fact_sales without additional credential setup
- ☐ Scheduling: visual workflow triggers via EventBridge Scheduler at the configured time without manual intervention
- ☐ QuickSight integration: dashboard launched from the project appears as a project asset and is shareable only with domain-registered groups

8. Success Criteria Checklist

8a. Infrastructure build — success criteria

Reduced scope per Section 1.

- ☐ Domain and project are created successfully via quick setup and are reachable by all attendees
- ☐ All three zone S3 buckets (raw, curated, rejected) block public access
- ☐ IAM Identity Center integration is functional for both SageMaker domain login and QuickSight (if in scope)
- ☐ The default Glue ETL compute connection works for Visual ETL flows (fine-grained mode not configured for this prototype)

8b. Data engineering build — success criteria

- ☐ The fractional Quantity pattern is investigated and documented, not ignored
- ☐ fact_sales + dim_product + dim_branch are visible and correctly typed in the Glue Data Catalog
- ☐ All three tables are published to SageMaker Catalog (Task D12) and discoverable from a different project
- ☐ Published tables carry column-level business metadata and at least one glossary term (Task D13)
- ☐ At least 4 of the 7 negative test cases from Section 7 are run and documented
- ☐ All 3 business-question flows (Section 3) run successfully against real data
- ☐ Amazon Q is used to draft the Business Question 2 flow, and any discrepancy from a manual approach is explained
- ☐ At least one attendee successfully runs the notebook EDA task (Task D14) against fact_sales
- ☐ At least one flow is scheduled successfully via the unified scheduling experience
- ☐ At least one QuickSight dashboard is published to SageMaker Catalog and shared within the workshop's directory group
- ☐ Every teammate can explain why the star schema was chosen over the flat source, and can name at least one thing that changed (and one thing that didn't) moving from standalone Glue to SageMaker Unified Studio

9. List of Infrastructure Created

Reduced scope per Section 1: quick setup auto-creates the VPC, default roles, and project bucket, so those don't need to be separately designed or built.

Resource	Type	Notes
Auto-created VPC (via quick setup)	VPC	Not the hand-designed production template - acceptable for this prototype only

Auto-created default IAM roles (Domain Execution, Domain Service, Manage access, Query execution, Provisioning, User role policy)	IAM Roles	Quick setup defaults, not hand-scoped least-privilege roles - revisit before production or client handoff
quickbite-workshop-raw-<account-id>	S3 Bucket	Raw zone - source CSV under sales_raw/ prefix
quickbite-workshop-curated-<account-id>	S3 Bucket	Curated zone - fact_sales partitioned by city/month, dimension tables unpartitioned
quickbite-workshop-rejected-<account-id>	S3 Bucket	Rejected/quarantined records, including flagged fractional-Quantity rows
Auto-created project-artifact S3 bucket (via quick setup)	S3 Bucket	Required by SageMaker Unified Studio itself; created automatically
IAM Identity Center permission set (QuickBite-Workshop-Attendee)	IDC Permission Set	Project-contributor scoped
Directory group (quickbite-etl-team)	IDC Group	Attendee cohort group
SageMaker Unified Studio domain	SageMaker Domain	IDC-based (recommended), created via quick setup
SageMaker workshop project	SageMaker Project	One project per pair, or one shared project
Default Glue ETL compute connection (project.spark.compatibility)	Compute Connection	Auto-created at project creation; used exclusively for this prototype
AWS Glue data source (crawler on raw CSV)	Data Source	Registers the raw schema in the Glue Data Catalog
JupyterLab notebook environment	Project Compute	Used for Task D14 - Python 3 or Glue PySpark kernel
SageMaker Catalog entries (fact_sales, dim_product, dim_branch + glossary terms)	Catalog Assets	Published in Task D12, enriched in Task D13
Visual workflow (crawl, transform, DQ check, publish)	Orchestration Asset	Scheduled via EventBridge Scheduler

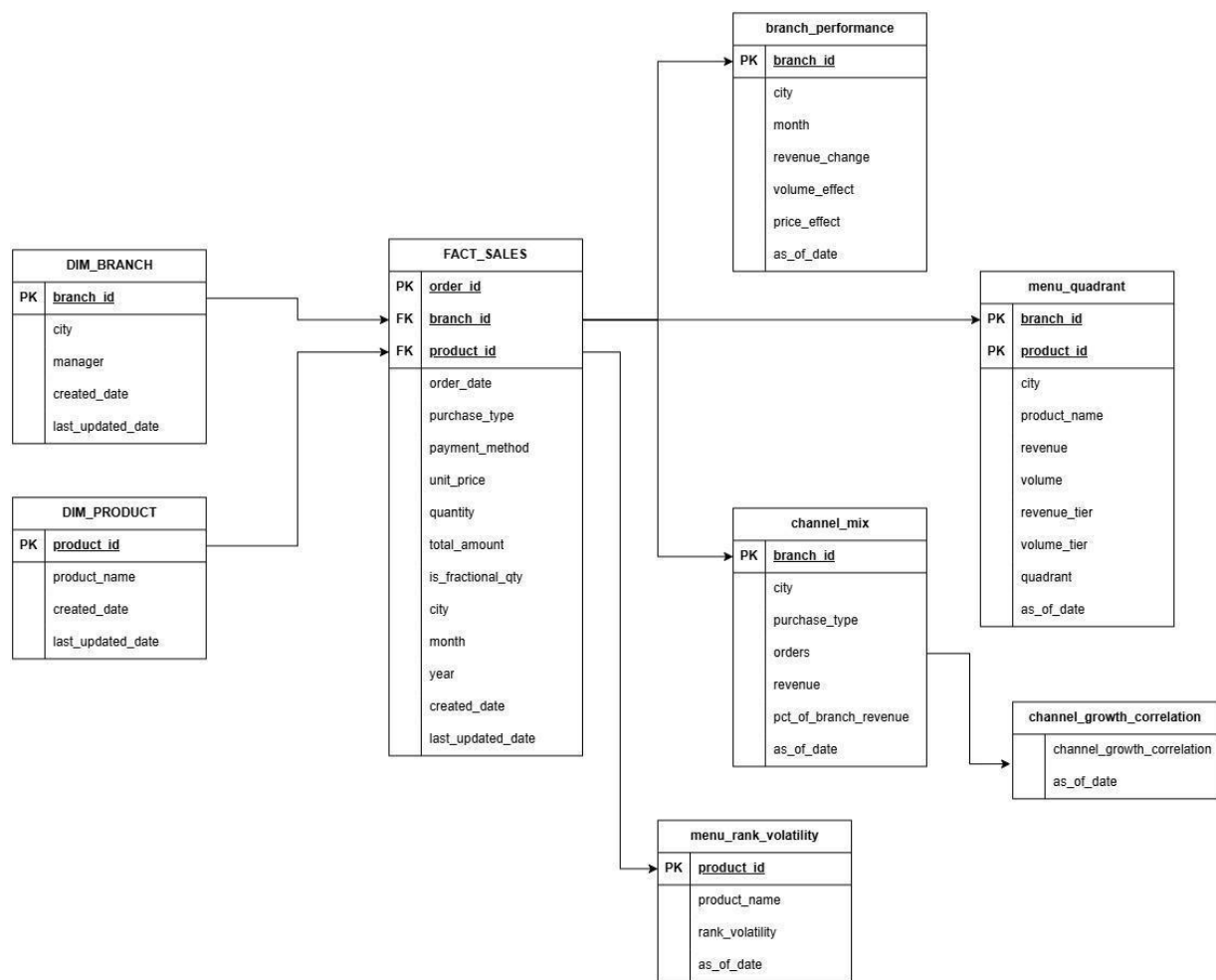
QuickSight dataset + dashboard	QuickSight Assets	Auto-created secured project folder; published to Catalog
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Appendix:

Visual ETL Output - Bronze to Silver



Data Model



Business Question 1 Analysis:



EXECUTIVE SUMMARY – Branch Performance: Price vs. Volume Analysis

OVERVIEW

Branches analysed : 4 (Berlin, Lisbon, London, Paris)
Periods with MoM : 9 branch-month transitions
Analysis window : 2022-11 → 2023-12

Total revenue change (all periods, all branches) : € -161,263.11
↳ Volume effect accounted for : € -94,177.32
↳ Price effect accounted for : € -67,085.79

BRANCH-LEVEL FINDINGS

Berlin	Net: ▼€	3,476		Vol: ▲	54,754		Price: ▼	58,229		Driver: Price-led
Lisbon	Net: ▼€	57,114		Vol: ▼	30,872		Price: ▼	26,242		Driver: Volume-led
London	Net: ▼€	76,474		Vol: ▼	94,427		Price: ▲	17,953		Driver: Volume-led
Paris	Net: ▼€	24,200		Vol: ▼	23,633		Price: ▼	567		Driver: Volume-led

KEY INSIGHTS

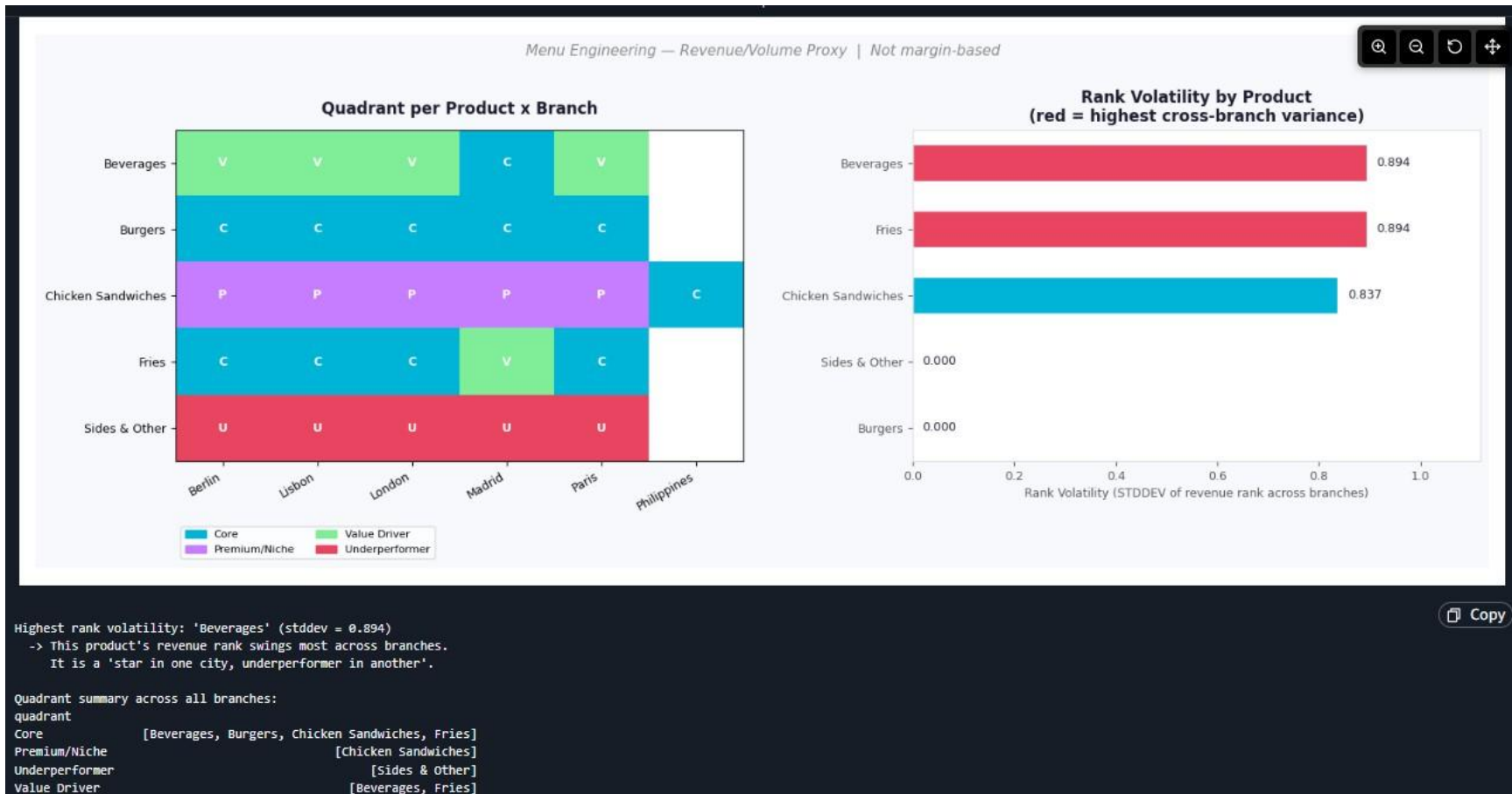
1. OVERALL DIRECTION: Revenue trend is volume-led across the network.
Volume effect (-94,177) vs Price effect (-67,086).
2. STRONGEST VOLUME PERFORMER: Berlin
→ Investigate footfall, transaction frequency, or promotional uplift.
3. STRONGEST PRICE PERFORMER: London
→ Evaluate pricing sustainability vs. volume erosion risk.
4. MOST AT-RISK BRANCH: London (highest cumulative revenue decline)
→ Audit pricing competitiveness and promotional activity.
5. SIGNAL PATTERNS TO WATCH:
 - Neg. volume + Pos. price → trading up but losing customers.
 - Pos. volume + Neg. price → discount-driven growth; margin at risk.
 - Both negative → structural decline requiring urgent review.

DATA QUALITY NOTES

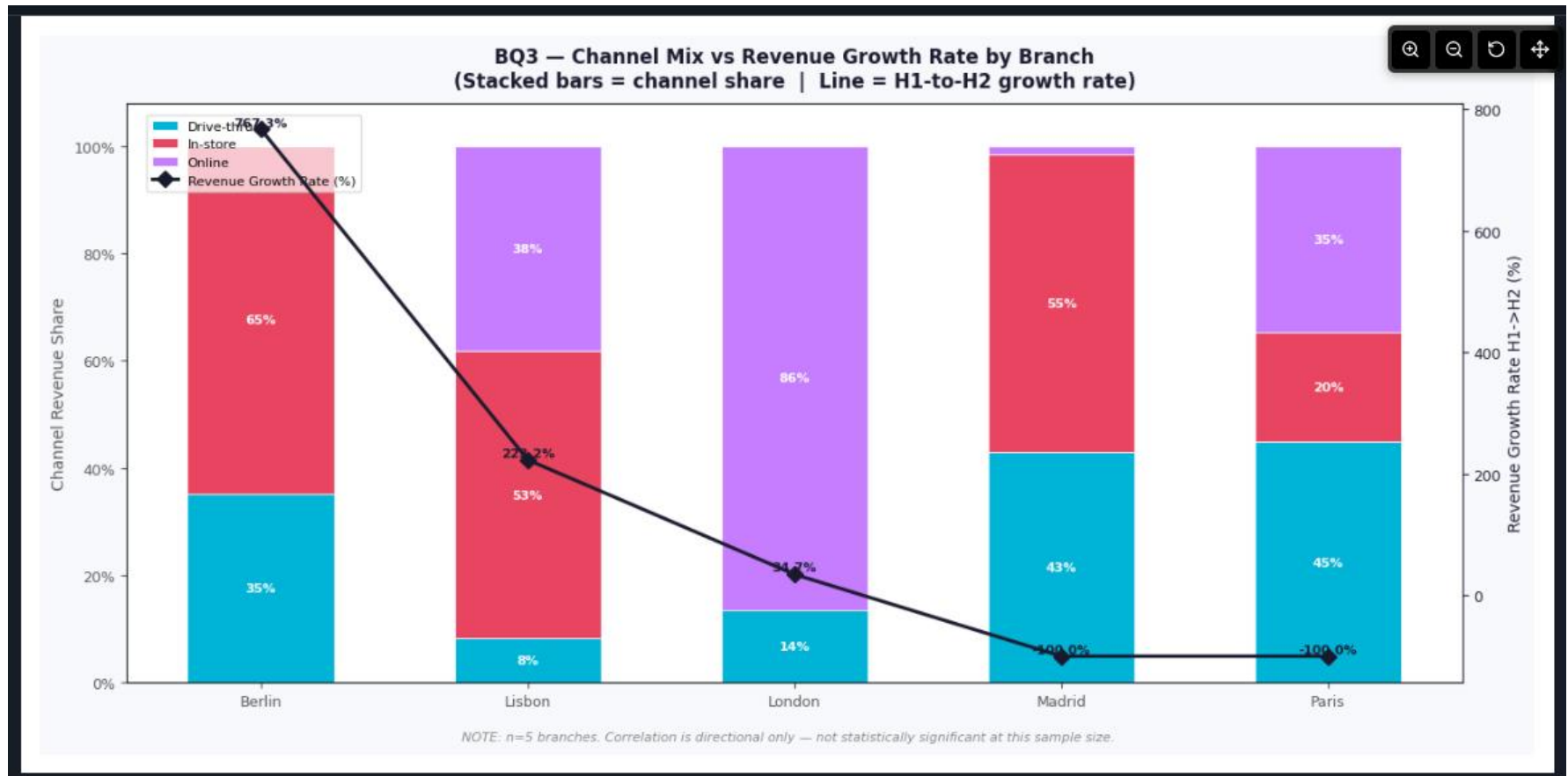
- Madrid (Nov-2022 only) and Philippines (Dec-2023 only) lack MoM comparison.
- 3 outlier dates (Sep/Dec-2023) may distort transitions – treat with caution.

All effects are approximations. Interaction term ($\Delta \text{qty} \times \Delta \text{price}$) omitted.

Business Question 2 Analysis:



Business Question 3 Analysis:



BQ3 – EXECUTIVE SUMMARY

Channel Behaviour: Does Channel Mix Track with Branch Performance?

ANALYSIS WINDOW : Nov 2022 – Dec 2022 (core data period)
BRANCHES COVERED : 5 (Berlin, Lisbon, London, Madrid, Paris)
CHANNELS OBSERVED: 3 (Drive-thru, In-store, Online)

CHANNEL MIX OVERVIEW

Each branch shows a distinct channel revenue composition.

Dominant channel per branch:

Berlin	-> In-store (64.8%)
Lisbon	-> In-store (53.3%)
London	-> Online (86.5%)
Madrid	-> In-store (55.4%)
Paris	-> Drive-thru (44.9%)

REVENUE GROWTH (H1 to H2, Nov-Dec 2022)

Highest growth : Berlin (767.3%)
Lowest growth : Madrid (-100.0%)

CHANNEL-GROWTH CORRELATION

Correlation (Online share x revenue growth rate): -0.3670
Direction : NEGATIVE
Strength : MODERATE (by conventional thresholds)

IMPORTANT CAVEAT:

This result is based on only 5 branches, which is far below the minimum sample size (~30) required for a statistically meaningful Pearson correlation. The result is illustrative of the analytical method only and MUST NOT be used for strategic decision-making until the branch network reaches 50+ locations.

DIRECTIONAL OBSERVATIONS

- A negative association (-0.37) suggests that branches with a higher share of "Online" revenue tend to show slower H2 growth – but this is a pattern to watch, not a conclusion to act on.
- Channel mix appears relatively stable across branches, with limited variation in dominant channel share – suggesting that no single branch has made a significant shift toward

References

- [Restaurant Sales Data \(rohitgrewal\) — Kaggle](#)
- [Visual ETL — Amazon SageMaker Unified Studio \(docs\)](#)
- [Glue ETL in Amazon SageMaker Unified Studio](#)
- [Configuring permission mode for Glue ETL](#)
- [Amazon SageMaker Unified Studio now supports AWS Glue 5.1](#)
- [Using visual workflows in IAM-based domains](#)
- [Amazon SageMaker scheduling experience for Visual ETL and Query editors](#)
- [Amazon QuickSight in Amazon SageMaker Unified Studio](#)
- [Create a SageMaker Unified Studio domain — quick setup](#)
- [Amazon S3 data in Amazon SageMaker Unified Studio](#)
- [Publish assets to the Amazon SageMaker Unified Studio catalog from the project inventory](#)